

41. Option (c) is correct.

$$\begin{aligned}\frac{\cos A + \cos B}{\cos\left(\frac{A-B}{2}\right)} &= \frac{2\cos\frac{A+B}{2} \cdot \cos\frac{A-B}{2}}{\cos\left(\frac{A-B}{2}\right)} \\ &= 2\cos\left(\frac{\pi-C}{2}\right) \\ &= 2\cos\left(\frac{\pi}{2} - \frac{C}{2}\right) \\ &= 2\sin\frac{C}{2} \\ &= 2\sin 30^\circ \\ &= 2 \times \frac{1}{2} \\ &= 1\end{aligned}$$

42. Option (c) is correct.

$$\begin{aligned}2 \cot^{-1} 3 &= 2 \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{\frac{2}{3}}{1 - \frac{1}{9}} \\ &= \tan^{-1} \left(\frac{2}{3} \times \frac{9}{8} \right) = \tan^{-1} \frac{3}{4} \\ \text{So, } \sqrt{15 + \cot^2 \left(\frac{\pi}{4} - 2 \cot^{-1} 3 \right)} \\ &= \sqrt{15 + \cot^2 \left(\tan^{-1} 1 - \tan^{-1} \frac{3}{4} \right)} \\ &= \sqrt{15 + \cot^2 \left[\tan^{-1} \left(\frac{1 - \frac{3}{4}}{1 + \frac{3}{4}} \right) \right]} \\ &= \sqrt{15 + \cot^2 \left(\tan^{-1} \frac{1}{7} \right)} \\ &= \sqrt{15 + \cot^2 (\cot^{-1} 7)} \\ &= \sqrt{15 + 49} \\ &= 8\end{aligned}$$

43. Option (b) is correct.

$$\begin{aligned}&\frac{\sin 10^\circ \cdot \sin 50^\circ + \sin 50^\circ \cdot \sin 250^\circ + \sin 250^\circ \cdot \sin 10^\circ}{\cos 240^\circ - \cos 260^\circ} \\ &= \frac{1}{2} (\cos 40^\circ - \cos 60^\circ + \cos 200^\circ - \cos 300^\circ + \\ &\quad \cos 240^\circ - \cos 260^\circ) \\ &= \frac{1}{2} \left[\cos 40^\circ - \frac{1}{2} + \cos(180+20)^\circ - \cos(360-60)^\circ \right. \\ &\quad \left. + \cos(180+60)^\circ - \cos(180+80)^\circ \right]\end{aligned}$$

$$\begin{aligned}&= \frac{1}{2} \left[\cos 40^\circ - \frac{1}{2} - \cos 20^\circ - \cos 60^\circ - \cos 60^\circ + \cos 80^\circ \right] \\ &= \frac{1}{2} \left[\cos 40^\circ - \cos 20^\circ + \cos 80^\circ - \frac{1}{2} - \frac{1}{2} - \frac{1}{2} \right] \\ &= \frac{1}{2} \left[-2 \sin 30^\circ \cdot \sin 10^\circ + \cos(90^\circ - 10^\circ) - \frac{3}{2} \right] \\ &= \frac{1}{2} \left[-\sin 10^\circ + \sin 10^\circ - \frac{3}{2} \right] \\ &= -\frac{3}{4}\end{aligned}$$

44. Option (b) is correct.

$$\begin{aligned}&\tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1} \left(\frac{a-b}{a+b} \right) \\ &= \tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1} \left(\frac{\frac{a}{b} - 1}{1 + \frac{a}{b} \cdot 1} \right) \\ &= \tan^{-1} \frac{a}{b} - \tan^{-1} \frac{a}{b} + \tan^{-1} 1 \\ &= \frac{\pi}{4}\end{aligned}$$

45. Option (b) is correct.

For real roots

$$(\cos \beta)^2 - 4 \sin \beta (\cos \beta - 1) \geq 0$$

Since $-1 \leq \cos \beta \leq 1$ and $\sin \beta \geq 0$ for $\beta \in \{0, \pi\}$

So, it is only possible when, $\cos \beta - 1 \geq 0$

46. Option (c) is correct.

$$\begin{aligned}&\cos 2A + \cos 2B + \cos 2C \\ &= 1 - 2\sin^2 A + 1 - 2\sin^2 B + 1 - 2\sin^2 C \\ &= 3 - 2(\sin^2 A + \sin^2 B + \sin^2 C) \\ &= 3 - 2 \left[\left(\frac{16}{65} \right)^2 + \left(\frac{63}{65} \right)^2 + 0 \right] = 1\end{aligned}$$

[Since, $16^2 + 63^2 = 65^2$]

47. Option (b) is correct.

$$\therefore \alpha + \beta = \frac{5\pi}{4}$$

$$\tan(\alpha + \beta) = \tan \frac{5\pi}{4}$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = \tan \left(\pi + \frac{\pi}{4} \right) = \tan \frac{\pi}{4}$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = 1$$

$$\tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta = 1$$